

Circles

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **identify and name** radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

RELATED SKILLS

I Identify circle parts and relationships from a diagram.

A Read and write geometric notation correctly.

- ☐ 2. I can **calculate** circumference and arc length.

RELATED SKILLS

I Calculate arc length from radius and central angle.

I Calculate circumference from radius or diameter.

I Relate a central angle to a fraction of a circle.

A Select and report appropriate linear, square, or angular units.

- ☐ 3. I can **calculate** the area of a circle, sector, or segment.

RELATED SKILLS

I Calculate sector area from radius and central angle.

I Calculate segment area by subtracting a triangle from a sector.

I Calculate the area of a circle.

A Select and report appropriate linear, square, or angular units.

- ☐ 4. I can **determine** angle and arc measures involving central and inscribed angles.

RELATED SKILLS

I Relate an inscribed angle to its intercepted arc.

D Relate a central angle to a fraction of a circle.

A Use inverse operations to maintain an equivalent equation.

- ☐ 5. I can **apply** relationships among chords, arcs, and distances from the center.

RELATED SKILLS

I Apply equal-chord, equal-arc, and center-distance relationships.

A Identify circle parts and relationships from a diagram.

- ☐ 6. I can **apply** tangent-radius and tangent-segment relationships.

RELATED SKILLS

I Select and apply an intersecting-segment product relationship.

I Use the perpendicular relationship between a radius and tangent.

☐ **7. I can **determine** angle measures formed by chords, secants, or tangents.**

RELATED SKILLS

☐ I Select the correct angle formula for chords, secants, or tangents.

☐ A Identify circle parts and relationships from a diagram.

☐ **8. I can **determine** segment lengths formed by intersecting chords.**

RELATED SKILLS

☐ I Select and apply an intersecting-segment product relationship.

☐ A Use inverse operations to maintain an equivalent equation.

☐ **9. I can **determine** segment lengths formed by secants or tangents.**

RELATED SKILLS

☐ D Select and apply an intersecting-segment product relationship.

☐ A Use inverse operations to maintain an equivalent equation.

☐ **10. I can **solve** a multi-step circle problem and justify each relationship used.**

RELATED SKILLS

☐ D Select a theorem that connects known facts to a target statement.

☐ A Identify circle parts and relationships from a diagram.

☐ A Select and apply an intersecting-segment product relationship.

☐ A Select the correct angle formula for chords, secants, or tangents.
